Decimal → Octal

Division-Remainder Method

Learn how to convert decimal numbers into octal using repeated division by 8.



How to Convert Decimal to Octal

Divide the decimal number by 8.

Record the remainder.

Repeat with the quotient until it reaches zero.

Read the remainders from bottom to top to get the octal number.



Example: Convert 83 to Octal

83 ÷ 8 = 10	remainder	3
↓		
10 ÷ 8 = 1	remainder	2
↓		
1 ÷ 8 = 0	remainder	1

Read remainders bottom → top: **1 2 3**

So, 83_{10} = 123_8



Your Turn! Convert 5010 to Octal

Try it yourself! You have 20 seconds.



Solution

:

50 ÷ 8 = 6 remainder 2

↓

6 ÷ 8 = 0 remainder 6

Read bottom → top: 6 2

5010 = 62$_8$

Recap: Decimal → Octal

1. Divide decimal number by 8 repeatedly.
2. Record remainders each time.
3. Stop when quotient is zero.
4. Read remainders from bottom to top for octal result.

